

A CERTIFIED $\mathcal{RM}(7, 3, 2)$ MIXED RADIAL MOORE GRAPH

ABSTRACT. We give an explicit mixed graph on 104 vertices that is totally $(7, 3)$ -regular and has radius 2 and diameter 3. Hence an $\mathcal{RM}(7, 3, 2)$ mixed radial Moore graph exists. The graph has a unique central vertex and status 1-norm $N_1 = 3378$. We also correct a single transposed table cell: the value $N_1 = 18$, correctly attributed in the text of [4] to the order-108 graphs with $(r, z) = (3, 7)$, is printed in [4, Table 3] and [1, Table 4.2] in the cell $(r, z) = (7, 3)$, and [1, p. 88] consequently reports the existence of $\mathcal{RM}(7, 3, 2)$ as guaranteed. The complete adjacency data of the graph are printed in Section 1.

1. THE CONSTRUCTION

A simple mixed graph $G = (V, E, A)$ has undirected edges

$$E \subseteq \binom{V}{2}$$

and directed arcs

$$A \subseteq \{(u, v) \in V^2 : u \neq v\},$$

with no directed digons and no edge–arc conflicts. Distances respect arc orientation, and all eccentricities below are out-eccentricities. The graph is *totally* (r, z) -regular if every vertex has undirected degree r and directed in-degree and out-degree z .

For diameter two, the mixed Moore bound is

$$(1) \quad M(r, z, 2) = 1 + (r + z) + r(r - 1 + z) + z(r + z) = (r + z)^2 + z + 1.$$

This is the diameter-two mixed Moore bound of Bosák [2]. A totally (r, z) -regular mixed graph of order $M(r, z, k)$, radius k , and diameter $k + 1$ is an $\mathcal{RM}(r, z, k)$ graph [4, 1].

For a vertex v , let

$$\sigma(v) = \sum_{u \in V} \text{dist}(v, u).$$

For $(r, z, k) = (7, 3, 2)$, the Moore status is

$$\sigma_M = 10 + 2 \cdot 93 = 196.$$

Following [4], define

$$N_1(G) = \|(\sigma(v))_{v \in V} - 196\mathbf{1}\|_1.$$

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Theorem 1. *There is an $\mathcal{RM}(7, 3, 2)$ graph on 104 vertices with a unique central vertex and*

$$N_1(G) = 3378.$$

Proof. The adjacency certificate below gives, for every vertex, its undirected neighbours $U(v)$ and the heads $A^+(v)$ of its outgoing arcs, and so determines a mixed graph G on

$$V = \{0, \dots, 103\}.$$

Undirected adjacency is symmetric there, and there are no loops, repeated elements, directed digons, or edge–arc conflicts. Every row lists seven undirected neighbours and three arc heads, and every vertex occurs as an arc head exactly three times. Hence

$$|E| = 364 = \frac{104 \cdot 7}{2}, \quad |A| = 312 = 104 \cdot 3,$$

and, at every vertex,

$$d_E(v) = 7, \quad d_A^+(v) = d_A^-(v) = 3.$$

Breadth-first search from every vertex gives

$$\{\text{ecc}(v) : v \in V\} = \{2^1, 3^{103}\}.$$

Thus G has radius 2, diameter 3, and a unique central vertex. The central distance-layer profile is

$$[1, 10, 93].$$

Since

$$M(7, 3, 2) = 10^2 + 3 + 1 = 104,$$

this proves that G is an $\mathcal{RM}(7, 3, 2)$ graph.

It remains to compute the status norm. For $v \in V$, let

$$q(v) = |\{u \in V : \text{dist}(v, u) = 3\}|.$$

Every vertex has exactly ten vertices at distance one, and all remaining vertices are at distance two or three. Hence

$$\sigma(v) = 10 + 2(93 - q(v)) + 3q(v) = 196 + q(v).$$

The total status is

$$\sum_{v \in V} \sigma(v) = 23762.$$

Therefore

$$N_1(G) = 23762 - 104 \cdot 196 = 3378.$$

Equivalently, exactly 3378 ordered pairs of vertices are at distance three. $\square$

Verification. The structural conditions, the all-pairs distances, the eccentricities, the statuses, and N_1 are finite computations on the 104 printed rows, and we performed them by computer directly from that table in exact integer arithmetic.

Adjacency certificate. For each vertex v , the table gives the undirected neighbours $U(v)$ and the heads $A^+(v)$ of the outgoing arcs. Vertex 0 is the central vertex; vertices $1, \dots, 7$ are its undirected neighbours and $8, 9, 10$ the heads of its arcs.

v	$U(v)$	$A^+(v)$	v	$U(v)$	$A^+(v)$
0	1, 2, 3, 4, 5, 6, 7	8, 9, 10	52	5, 15, 33, 41, 51, 72, 91	35, 43, 90
1	0, 11, 12, 13, 14, 15, 16	17, 18, 19	53	20, 43, 57, 61, 74, 84, 101	30, 59, 98
2	0, 20, 21, 22, 23, 24, 25	26, 27, 28	54	36, 49, 55, 64, 65, 66, 97	25, 31, 85
3	0, 29, 30, 31, 32, 33, 34	35, 36, 37	55	27, 38, 54, 72, 85, 100, 102	22, 80, 96
4	0, 38, 39, 40, 41, 42, 43	44, 45, 46	56	6, 11, 51, 61, 66, 68, 81	7, 46, 88
5	0, 47, 48, 49, 50, 51, 52	53, 54, 55	57	6, 16, 27, 38, 46, 53, 84	2, 71, 96
6	0, 56, 57, 58, 59, 60, 61	62, 63, 64	58	6, 42, 47, 59, 70, 88, 94	15, 71, 86
7	0, 65, 66, 67, 68, 69, 70	71, 72, 73	59	6, 15, 29, 31, 58, 68, 90	44, 82, 91
8	74, 75, 76, 77, 78, 79, 80	81, 82, 83	60	6, 21, 25, 31, 34, 37, 75	14, 49, 51
9	84, 85, 86, 87, 88, 89, 90	91, 92, 93	61	6, 22, 39, 47, 53, 56, 75	2, 55, 83
10	94, 95, 96, 97, 98, 99, 100	101, 102, 103	62	12, 23, 25, 28, 36, 63, 85	11, 48, 74
11	1, 26, 32, 40, 56, 86, 93	2, 15, 87	63	28, 29, 32, 62, 70, 98, 101	15, 39, 56
12	1, 16, 18, 62, 83, 90, 100	36, 74, 103	64	13, 39, 54, 72, 82, 95, 101	77, 79, 88
13	1, 16, 23, 64, 69, 73, 79	5, 92, 93	65	7, 46, 51, 54, 72, 84, 96	8, 77, 97
14	1, 17, 28, 46, 67, 80, 98	7, 57, 66	66	7, 54, 56, 68, 74, 75, 96	6, 29, 37
15	1, 22, 39, 46, 48, 52, 59	6, 30, 61	67	7, 14, 25, 26, 27, 40, 77	11, 87, 100
16	1, 12, 13, 28, 33, 57, 95	23, 40, 61	68	7, 21, 56, 59, 66, 86, 103	47, 52, 65
17	14, 32, 71, 76, 80, 95, 96	28, 34, 53	69	7, 13, 20, 77, 82, 93, 98	3, 12, 27
18	12, 27, 33, 38, 71, 73, 99	57, 64, 70	70	7, 34, 37, 42, 43, 58, 63	41, 92, 93
19	26, 27, 36, 77, 81, 87, 91	11, 25, 34	71	17, 18, 28, 35, 44, 72, 73	0, 19, 46
20	2, 44, 49, 53, 69, 74, 75	21, 24, 99	72	52, 55, 64, 65, 71, 88, 89	10, 70, 75
21	2, 35, 60, 68, 79, 88, 102	29, 31, 94	73	13, 18, 22, 45, 51, 71, 83	9, 21, 32
22	2, 15, 23, 37, 61, 73, 84	13, 86, 97	74	8, 20, 34, 53, 66, 94, 99	32, 84, 96
23	2, 13, 22, 29, 42, 62, 89	41, 43, 78	75	8, 20, 60, 61, 66, 92, 98	9, 59, 82
24	2, 33, 34, 41, 48, 76, 86	54, 80, 101	76	8, 17, 24, 25, 42, 51, 90	14, 69, 85
25	2, 37, 45, 60, 62, 67, 76	4, 35, 63	77	8, 19, 38, 47, 67, 69, 92	0, 18, 32
26	11, 19, 30, 36, 67, 78, 85	4, 17, 24	78	8, 26, 40, 44, 80, 92, 96	10, 43, 79
27	18, 19, 45, 46, 55, 57, 67	24, 75, 90	79	8, 13, 21, 29, 35, 89, 93	19, 37, 81
28	14, 16, 48, 62, 63, 71, 82	50, 60, 72	80	8, 14, 17, 35, 78, 83, 93	34, 51, 99
29	3, 23, 59, 63, 79, 90, 98	67, 84, 95	81	19, 36, 56, 87, 89, 97, 102	17, 38, 68
30	3, 26, 32, 44, 47, 84, 90	16, 60, 100	82	28, 34, 45, 48, 49, 64, 69	18, 30, 56
31	3, 45, 50, 59, 60, 87, 96	13, 33, 42	83	12, 41, 73, 80, 94, 95, 101	58, 63, 76
32	3, 11, 17, 30, 45, 63, 86	12, 23, 57	84	9, 22, 30, 53, 57, 65, 97	21, 80, 85
33	3, 16, 18, 24, 34, 38, 52	25, 26, 79	85	9, 26, 41, 49, 55, 62, 102	28, 38, 68
34	3, 24, 33, 60, 70, 74, 82	1, 26, 39	86	9, 11, 24, 32, 40, 50, 68	29, 98, 99
35	21, 39, 42, 44, 71, 79, 80	77, 78, 83	87	9, 19, 31, 45, 81, 89, 94	20, 47, 56
36	19, 26, 54, 62, 81, 91, 103	7, 67, 102	88	9, 21, 58, 72, 92, 94, 103	6, 38, 76
37	22, 25, 60, 70, 93, 101, 103	49, 89, 100	89	9, 23, 51, 72, 79, 81, 87	42, 74, 97
38	4, 18, 33, 48, 55, 57, 77	58, 67, 73	90	9, 12, 29, 30, 48, 59, 76	20, 50, 70
39	4, 15, 35, 43, 61, 64, 103	3, 62, 94	91	19, 36, 41, 42, 47, 52, 99	13, 90, 95
40	4, 11, 49, 50, 67, 78, 86	1, 47, 81	92	50, 75, 77, 78, 88, 98, 101	0, 45, 58
41	4, 24, 52, 83, 85, 91, 95	8, 16, 102	93	11, 37, 43, 69, 79, 80, 97	39, 55, 66
42	4, 23, 35, 58, 70, 76, 91	33, 51, 98	94	10, 58, 74, 83, 87, 88, 103	4, 23, 91
43	4, 39, 53, 70, 93, 99, 100	48, 54, 101	95	10, 16, 17, 41, 46, 64, 83	48, 52, 59
44	20, 30, 35, 71, 78, 97, 102	40, 65, 68	96	10, 17, 31, 65, 66, 78, 99	5, 22, 73
45	25, 27, 31, 32, 73, 82, 87	16, 20, 89	97	10, 44, 50, 54, 81, 84, 93	5, 14, 95
46	14, 15, 27, 57, 65, 95, 102	62, 69, 94	98	10, 14, 29, 63, 69, 75, 92	12, 65, 66
47	5, 30, 58, 61, 77, 91, 100	41, 42, 52	99	10, 18, 43, 49, 74, 91, 96	40, 72, 87
48	5, 15, 24, 28, 38, 82, 90	69, 76, 84	100	10, 12, 43, 47, 50, 55, 103	44, 53, 60
49	5, 20, 40, 54, 82, 85, 99	3, 31, 64	101	37, 53, 63, 64, 83, 92, 102	33, 36, 103
50	5, 31, 40, 86, 92, 97, 100	22, 61, 89	102	21, 44, 46, 55, 81, 85, 101	45, 50, 88
51	5, 52, 56, 65, 73, 76, 89	27, 49, 75	103	36, 37, 39, 68, 88, 94, 100	1, 78, 86

2. A PARAMETER CORRECTION

Jørgensen constructed two nonisomorphic mixed Moore graphs of order 108 with undirected degree 3 and directed degree 7 [5]. Ceresuela, López, and Chemisana applied an arc swap to these graphs and obtained the status vector

$$204^{98}, 205^8, 209^2,$$

which has status 1-norm

$$8(205 - 204) + 2(209 - 204) = 18.$$

Their text identifies the underlying Moore status vector as

$$\mathbf{s}_{3,7,2} = 204^{108},$$

but Table 3 places the value 18 in the cell $(r, z) = (7, 3)$ [4, Table 3]. Abiad, Ceresuela, and López subsequently state that the existence of $\mathcal{RM}(7, 3, 2)$ is guaranteed and repeat the same assignment in [1, p. 88 and Table 4.2]. This label is contested in print: Ceresuela’s thesis states that Jørgensen’s order-108 graphs have $r = 7$, $z = 3$, while it reproduces the status vector as $\mathbf{s}_{3,7,2}$ [3].

The order determines the correct parameter pair in these cases. Put

$$s = r + z.$$

By (1),

$$s^2 + 1 \leq M(r, z, 2) \leq s^2 + s + 1.$$

If $M(r, z, 2) = 108$, then $s = 10$, and therefore

$$z = 108 - 10^2 - 1 = 7, \quad r = 3.$$

If $M(r, z, 2) = 104$, then again $s = 10$, but

$$z = 104 - 10^2 - 1 = 3, \quad r = 7.$$

Consequently, the published value $N_1 = 18$ comes from an $\mathcal{RM}(3, 7, 2)$ graph and does not provide a certificate for the transposed parameter pair $\mathcal{RM}(7, 3, 2)$. Theorem 1 supplies such a certificate.

Remark 2. The correction concerns the provenance of the entry 18. It does not show that $N_1 = 18$ is impossible for an $\mathcal{RM}(7, 3, 2)$ graph. No optimality claim is made for the value 3378.

DATA AVAILABILITY

All data required to verify Theorem 1 are printed in the adjacency certificate in Section 1. A verification program that recomputes the assertions of Theorem 1 from that printed table accompanies this paper; it reads no further data.

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